MILITARY SPECIFICATION

# KETTLES, HEATING, BITUMEN, JOINT-SEALER TRAILER MOUNTED

This specification is approved for use by all Departments and Agencies of the Department of Defense.

## 1. SCOPE

1.1 Scope. This specification covers trailer-mounted kettles for heating and applying joint-sealing compound conforming to SS-S-1401 and SS-S-1614.

1.2 Classification. The kettles shall be of the following types and sizes, as specified (see 6.2):

Type I - Liquefied petroleum gas (LPG) as burner and engine fuel.
Type II - Kerosene (light oil) as burner fuel and gasoline as engine fuel.
Type III - LPG as burner fuel and gasoline as engine fuel.
Type IV - LPG as burner fuel and without engine.

Size 75 - Melting tank capacity of 75 gallons.
Size 120 - Melting tank capacity of 120 gallons.
Size 165 - Melting tank capacity of 165 gallons.
Size 200 - Melting tank capacity of 200 gallons.

## 2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specifications, standards, and handbooks. Unless otherwise specified, the following specifications, standards, and handbooks of the issue listed in that issue of the Department of Defense Index of Specifications and Standards (DODISS) specified in the solicitation, form a part of this specification to the extent specified herein.

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: HQ AFLC/CASO/LODSAC, Federal Center, Battle Creek, MI 49016, by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

FSC 3895

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SPECIFICATIONS

FEDERAL

HH-I-558 - Insulation, BLocks, Boards, Blankets, Felts, Sleeving (Pipe and Tube Covering), and Pipe Fitting Covering, Thermal (Mineral Fiber, Industrial Type).
SS-S-1401 - Sealing Compound, Hot Applied, for Concrete and Asphalt Pavements.
SS-S-1614 - Sealing Compound, Jet-Fuel Resistant, Hot-Applied, One Component, for Portland Cement and Tar Concrete Pavements.

MILITARY

MIL-P-514 - Plates; Identification, Transportation Data, and Blank.
MIL-T-704 - Treatment and Painting of Materiel.
MIL-D-771 - Distributors, Bitumen and Water, Packaging of.
MIL-B=11188 - Batteries, Storage: Lead-Acid.
MIL-I-22023 - Insulation Felt, Thermal and Sound Absorbing, Fibrous Glass, Flexible..
MIL-B-23272 - Battery, Storage: Alkaline, Nickel-Cadmium.

STANDARDS

FEDERAL

FED STD H28 - Screw-Thread Standards for Federal Services.
FED STD No. 595 - Colors.

MILITARY

MIL-STD-130 - Identification Marking of US Military Property.
MS51336 - Lunette-Coupler, Drawbar, Ring.

(Copies of specifications, standards, handbooks, drawings, and publications required by manufacturers in connection with specific acquisition functions should be obtained from the contracting activity or as directed by the contracting officer.)

2.2 Other Government documents, drawings, and publications. The following other Government documents form a part of this specification to the extent specified herein.

DEPARTMENT OF TRANSPORTATION (DoT)

Motor Carrier Safety Regulations.

(Application for copies should be addressed to the Superintendent of Documents, Government Printing Office, Washington, DC 20402.)



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2.3 Order of precedence. In the event of a conflict between the text of this specification and the references cited herein, the text of this specification shall take precedence.

2.4 Other publications. The following document(s) form a part of this specification to the extent specified herein. The issues of the documents which are indicated as DoD adopted shall be the issue listed in the current DODISS and the supplement thereto, if applicable.

AMERICAN NATIONAL STANDARDS INSTITUTE, INC. (ANSI)

B29.1 - Precision Power Transmission Roller Chains, Attachments and Sprockets.

(Application for copies should be addresssed to the American National Standards Institute, Inc., 1430 Broadway, New York, NY 10018.)

SOCIETY OF AUTOMOTIVE ENGINEERS, INC. (SAE)

SAE Standards and Recommended Practices.

(Application for copies should be addressed to the Society of Automotive Engineers, Inc., 400 Commonwealth Drive, Warrendale, PA 15096).

TIRE AND RIM ASSOCIATION, INC.

T&RA Yearbook.

(Application for copies should be addressed to the Tire and Rim Association, Inc., 3200 West Market Street, Akron, OH 44313.)

UNDERWRITERS' LABORATORIES, INC. (UL)

Standard for Gas Cylinders.

(Application for copies should be addressed to the Underwriters' Laboratories, Inc., 333 Pfingsten Road, Northbrook, IL 60062.)

(Industry association specifications and standards are generally available for reference from libraries. They are also distributed among technical groups and using Federal agencies.)

3. REQUIREMENTS

3.1 Description. The kettle shall be of the indirect heating type consisting of a melting compartment, a heat-transfer oil compartment, a burner, and all necessary controls for the operation of the kettle. The kettle and all components shall be mounted on a pneumatic-tired, two-wheeled trailer.

3.2 First article. When specified in the contract or purchase order, (see 6.2), the contractor shall furnish the required type and size kettle for first article inspection and approval (see 4.2.1 and 6.2).



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3.3 Base of maintenance. The kettle shall permit accessibility to all parts for maintenance and repair utilizing conventional tools associated with kettles. The replacement and adjustment of parts shall be accomplished with minimum disturbance to other components.

3.4 Materials. Materials used shall be free from defects which would adversely affect the performance or maintainability of individual components or of the overall assembly. Materials not specified herein shall be of the same quality used for the intended purpose in commercial practice. Unless otherwise specified herein, all equipment, material, and articles incorporated in the work covered by this specification are to be new and fabricated using materials produced from recovered materials to the maximum extent possible without jeopardizing the intended use. The term "recovered materials" means materials which have been collected or recovered from solid waste and reprocessed to become a source of raw materials, as opposed to virgin raw materials. None of the above shall be interpreted to mean that the use of used or rebuilt products are allowed under this specification unless otherwise specified.

3.5 Common parts. Common hardware and fastening devices, such as bolts, nuts, screws, and washers, shall be held to a minimum number of types and sizes. All threads shall be in accordance with FED STD H28.

3.6 Safety. All exposed parts which are subject to high operating temperatures or which are energized electrically shall be insulated, fully enclosed, or guarded. All moving parts which are hazardous to operating or maintenance personnel shall be fully enclosed or guarded. Protective devices shall not impair the operating functions.

3.7 Performance. With the kettle stabilized at an ambient temperature of not more than 75° Fahrenheit (F), the temperature of the joint-sealing compound (charge of not less than 50 percent of the melting compartment capacity) shall be raised from 75°F to 425°F within a 5 hour period. After the temperature of the initial joint-sealing compound reaches 415°F, the kettle shall continue to heat additional joint-sealing compound to not less than 415°F at the following applicable rates:

Size 75 - 40 gallons per hour.
Size 120 - 60 gallons per hour.
Size 165 - 80 gallons per hour.
Size 200 - 100 gallons per hour.

3.8 Heat-transfer oil compartment. The heat-transfer oil compartment shall have steel walls not less than 3/16 inch thick. The capacity of the heat-transfer oil compartment shall be between 40 percent and 75 percent of the capacity of the melting compartment. A drain that will completely drain the heat-transfer oil compartment shall be provided. The filler opening shall be provided with a vent-type cap to prevent excessive pressure build-up within the compartment. A dipstick, graduated to indicate the maximum and minimum safe operating levels of the heat-transfer oil, shall be provided. A drain for draining condensation from the heat transfer storage tank shall be located in the tank bottom.



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3.9 Melting compartment. The melting compartment shall be formed by the inner wall of the heat transfer oil compartment and shall have a liquid capacity of not less than 4 percent greater than the applicable capacities listed in section 1.2 to provide for expansion of heated compound. The melting compartment shall be divided into a melting section and reservoir section (for melted material) by a removable separator. The separator shall be provided with openings to permit the material to flow into the reservoir section. All surfaces of the melting compartment shall be smooth and accessible for cleaning. A hinged cover shall be provided for the melting compartment. When fully opened the hinged cover shall remain in the open position by its own weight. The cover shall be of sufficient size to permit charging the melting compartment with joint-sealing compound in blocks or chunks up to 50-pound size. The opening shall be of such size that splashing of melted material will be at a minimum when tank is full and unit is towed at application speeds. A drain fitted with a lever operated or gate type shutoff valve shall be located at the lowest point of the melting compartment or connecting piping. All material piping lines shall be routed through the heat transfer oil compartment, or if external, shall be sufficiently insulated to prevent cooling of the material prior to entering the pump. Means for recirculating heated compound shall be provided.

3.10 Thermometers. Thermometers shall be provided to indicate the temperature of the heat-transfer oil and the sealing compound. The thermometers shall be located to permit reading from a position near the kettle outlet. Each thermometer shall be calibrated in increments of not more than 10°F.

3.11 Insulation. The outer shell of the oil compartment and the cover for the melting compartment shall be covered with insulation equal to or conforming to HH-I-558 or MIL-T-22023, type I, class 4.

3.12 Firebox and flue. The firebox and flues shall operate without preheating to obtain stack effect.

3.13 Protective cover. When specified (see 6.2), a protective cover shall be provided. The cover shall be weatherproof, flame resistant, and shall protect all components of the kettle.

3.14 Trailer. The kettle shall be mounted on a pneumatic-tired, 2-wheel, cart-type trailer chassis for towing over paved roads at speeds up to 25 miles per hour (mph). The frame shall withstand the static and dynamic loads imposed when the kettle is filled to capacity with sealing compound, heat transfer oil and fuel tank.

3.14.1 Lunette and safety chains. The tongue of the trailer shall be provided with a lunette conforming to MS51366. The melting compartment shall be level when the lunette is between 32 and 34 inches above the ground. Two safety chains conforming to DoT Motor Carrier Safety Regulations shall be attached to the tongue. Each chain shall extend between 36 and 42 inches beyond the lunette eye and shall be equipped with a safety hook. The widest dimension of the hook shall just pass through a 2-inch diameter opening.



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3.14.2 Support jacks. The trailer shall be equipped with two adjustable and retractable support jacks, one mounted at the front of the trailer and one mounted at the rear of the trailer. When the support jacks are retracted, the clearance between the jacks and the ground shall be not less than the clearance between the axle and the ground.

3.14.3 Suspension. The kettle shall be attached to the axle by semi-elliptical leaf springs. The axle and springs shall be of a rated capacity in excess of the load imposed when the kettle is filled to capacity with fuel, sealing compound, and heat-transfer oil.

3.14.4 Wheels and rims. The trailer wheels shall be the manufacturer's standard. Rim and tire ratings shall conform to the T&RA recommendations for the type and size of tires furnished.

3.14.5 Tires and tubes. Tires may be of the tube or tubeless type, shall be not less than 100 level quality, and shall have a load rating not less than the actual load imposed on the tire. Anti-ozone compounds shall be used to retard weathering of the tires and tubes.

3.14.6 Fenders. The kettle wheels shall be shielded by fenders to restrict water and other road-surface substances which may be thrown from the wheels. Each fender shall withstand a 200-pound load over any area of 10 inches square. Each fender shall have a full backplate to protect the tires from the heat radiated from the kettle.

3.15. Type I. The type I kettle shall operate on LPG as fuel for the engine and the burner.

3.15.1 Engine. The engine shall be the manufacturer's standard commercial engine and shall be equipped for burning LPG as fuel. The engine shall be equipped with an electrical cranking system, housing, a 5-ampere(minimum) battery-charging system, and instrument panel. As a minimum, the following instruments shall be provided:

a. Lubricating-oil pressure gage.
b. Indicator, cooling-liquid temperature (liquid-cooled engine).
c. An hourmeter.
d. Indicator, battery-charging.

3.15.2 Fuel tanks and controls. Provisions shall be made to mount two 100-pound size, LPG cylinders on the kettle. Means shall be provided to keep the cylinders in place. All controls that regulate the flow of gas shall be UL approved. Temperature control valve shall be in an enclosed metal box for protection. LPG cylinders shall not be furnished.

3.15.3 Burner(s). The burners shall be of either the retention flame type or the burner and pilot type and shall be designed for burning LPG as fuel. Burners shall be provided with a safety control device that will automatically shut off the fuel if the flame is extinguished or if excessive heat transfer oil temperture is reached. The burners shall be removable and shall be controlled by an adjustable thermostat. The thermostat shall automatically maintain the temperature of the heat transfer oil at any preset value, $\pm 10^{\circ}F$, between $100^{\circ}F$ and $550^{\circ}F$,. A manually-operated fuel shutoff valve shall be installed in the fuel line between the fuel tank and the burner.



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3.15.4 Pumping system. An engine-driven driven material pump for applying the joint sealing compound shall be installed on the kettle. The pumping system shall include a built-in pressure-relief valve, provisions for a pressure gage, piping, valves, and other controls that will permit the following functions:

a. Loading liquid material by drawing it from an outside source.
b. Circulate material by drawing it from the melting compartment and returning it thereto.
c. Pressure-apply melted sealing compound through the accessories specified herein.
d. Pump cleaning solvents through the system without the solvents entering the melting compartment.

3.15.5 Drive train. The drive train between the engine and the material pump shall be direct drive or include a hand-operated clutch of the plate or disk type. When used, the clutch shall transmit 150 percent of the maximum engine torque. Power transmission chains used in the drive train shall conform to ANSI B29.1. All components shall be accessible for lubrication, adjustment, and repair.

3.15.6 Discharge hose. A minimum 25-foot length of flexible metal hose having 1-inch inside diameter shall be provided. The hose shall be equipped with ground-joint unions that permit attaching the hose to the kettle and to the sealer bar.

3.15.7 Sealer bar. A hand-operated sealer bar shall be provided and shall be equipped with a valve for regulating the flow of the material through the sealer tips. Two handles that minimize heat transfer shall be provided on the bar.

3.15.8 Agitator. A removable, engine-driven agitator shall be installed in the melting compartment. The agitator shall be capable of agitating the material in both sections of the melting compartment when a second melting compartment is used (see 3.9).

3.15.9 Tool heating compartment. A tool heating compartment shall be provided to heat the hose, sealer bar, and sealer tips.

3.15.10 Battery(s) and battery box.

3.15.10.1 Battery(s). Unless otherwise specified (see 6.2), the battery(s) shall conform to MIL-B-11188 and shall be dry charged and sealed to prevent absorbtion of atmospheric moisture. Electrolyte shall be not be furnished. When specified (see 6.2), the battery(s) shall conform to MIL-B-23272.

3.15.10.3 Battery box. A weatherproof box made of steel 0.0747-inch nominal thickness shall be provided for the battery(s). Vents shall be provided near the top of the box and drains shall be provided in the bottom. The box shall be equipped with built-in restraints constructed so that pressure will be applied to the outer battery case only. When the batteries are installed in the box and with all electrical connections completed and



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when the battery box lid is in the closed position, there shall be no uninsulated electrical conductor within 1 inch of any point within the battery box at which an electrical potential exists. The box shall be located so that the battery(s) will be accessible for routine maintenance, removal, and replacement. The battery(s) shall be accessible for servicing with battery servicing aids such as a hydrometer and a syringe. All battery lead cables shall be protected to prevent chafing. All interior surfaces of the box shall be treated with acid-resistant paint.

3.16 Type II. The type II kettle shall be identical to the type I kettle, except the burners shall be fueled by kerosene and the engine shall conform to 3.16.1.

3.16.1 Engine. The engine shall be the manufacturer's standard commercial gasoline engine. Power and speed rating of the engine shall be such that operation of the equipment at rated conditions shall not require horsepower or speed in excess of the continuous brake horsepower rating of the engine at the applicable speed. The engine shall be equipped with a 12-volt electrical cranking system and shall be equipped with, but not limited to a fuel tank of a capacity for not less than 4 hours rated load operation, generator or alternator, engine housing, and instrument panel. The following instruments shall be provided:

a. Lubricating-oil pressure gage.
b. Indicator, cooling-liquid temperature (liquid-cooled-engine).
c. An hourmeter.
d. Indicator, battery-charging.

3.16.2 Burner fuel tank. The fuel tank shall have a capacity for not less than 8 hours of continuous operation. The filler opening shall be not less than 2-1/2 inches in diameter and the opening shall be fitted with a removable screen. The cap for the filler opening shall be attached by a chain.

3.16.3 Burner(s). The burners shall be removable and shall be controlled by an adjustable thermostat. The thermostat shall automatically maintain the temperature of the heat-transfer oil at any present value between 100°F and 550°F, within ±25°F. A manually-operated, shutoff valve shall be installed in the fuel line between the fuel tank and the burner.

3.17 Type III. The type III kettle shall be identical to the type I, except the engine shall conform to 3.16.1.

3.18 Type IV. The type IV kettle shall operate on LPG as fuel for the burners. An engine shall not be provided. Fuel tanks and controls shall conform to 3.15.2.

3.19 Lubrication. All surfaces that require lubrication shall be provided with an accessible means for lubricating.

3.19.1 Lubricants. The kettle shall operate as specified herein when lubricated with standard lubricants.



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3.19.2 Lubrication fittings. Fittings shall be located in a protected position and shall be accessible to a grease gun equipped with a straight hydraulic coupler. Accessibility to fittings shall be provided without the removal or adjustment of accessories or parts. Panels and plates equipped with hand-operable, quick-disconnect fasteners may be removed to provide accessibility.

3.19.3 Pressure-release device. An automatic pressure-release device shall be provided where the use of pressure lubricating equipment will damage grease seals or other parts.

3.19.4 Filling, draining, and checking provisions. Enclosures such as gear cases and transmission housings that contain a reservoir of lubricants shall be equipped with dipsticks, check plugs not less than 1/2-inch pipe size, or sight gages, to determine the level of the lubricant. Each enclosure shall be equipped for filling the enclosure with lubricant and for draining. The drain shall be fitted with a magnetic drain plug. The drain outlet shall be located so that removal of the plug will result in complete drainage of the lubricant from the enclosure. Drainage shall be to the ground when the kettle is in its normal position. Integral tubes or troughs may be used to convey the lubricant from the drain outlet to the ground.

3.19.5 Grease lubrication. All grease lubrications, including lubrication of sealed bearings, shall be as specified in section 5. The kettle shall be assembled, run-in, tested, and delivered with these greases. The bearings shall be cleaned prior to lubrication. Each compartment shall be tagged in a conspicuous place to indicate which grease has been used.

3.19.6 Oil lubrication. Unless otherwise preserved for packaging, all components shall be filled to operating level with winter grades of oils designed for use in the temperature range of 0°F to 32°F. Each component shall be tagged in a conspicuous place to indicate the temperature range and grade of lubricants used.

3.20 Lifting attachments. When specified (see 6.2), permanently affixed attachments that enable the kettle to be lifted in its normal travel position shall be provided. Each attachment shall withstand, without damage to any part of the kettle, 2-1/2 times the resultant vectorial force imposed when the kettle is suspended by the lifting attachment. Attachments for multiple slings shall be located so that: (a) slings will converge at not more than 19 feet above the lowest part of the kettle, and (b) not less than 1-inch clearance will be maintained between the slings and the kettle. The inside diameter of the attachment eye shall be not less than 3 inches. Lifting attachments may also be used as tiedown attachments when such attachments meet the requirements specified in 3.21.

3.21 Tiedown attachments. When specified (see 6.2), permanently affixed attachments that permit fastening the kettle to the floor or deck of a transportation medium shall be provided. Each attachment shall withstand, without damage to the kettle, its proportion of the following loads: 8-G forward, 8-G rearward, 2-1/4 G upward, 1-1/2 G sideward, where G equals cargo weight. Attachments shall be rounded and beveled to prevent cutting of the tiedown material. Tiedown attachments may also be used as lifting attachments when such attachments meet the requirements specified in 3.20.



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3.22 Identification marking. The kettle shall be identified in accordance with MIL-STD-130. The identification and transportation data plates shall conform to MIL-P-514. Unless otherwise specified (see 6.2), the material for the plates shall be photosensitive anodized aluminum. The transportation data plate shall indicate the location of the lifting and tiedown attachments.

3.23 Stenciling. When specified (see 6.2), the gross weight of each unit shall be stenciled on each side of the unit in such a manner as to be discernible. The prescribed tire pressure shall be stenciled near each wheel using letters not less than 1 inch high. Stenciling shall conform to MIL-T-704.

3.24 Technical publications. When specified (see 6.2), commercial technical publications shall be furnished. Commercial technical publications are operators manual, service manual, and repair/parts manual.

3.25 Instruction plate. An instruction plate conforming to MIL-P-514, type IX, shall be provided and shall be affixed to the kettle in a conspicuous place. The plate shall contain clear and concise instructions for introducing cleaning solvents into the pumping system and discharge hose. In addition, a warning plate of the same material as the instruction plate, with red lettering in accordance with FED STD No. 595, color No. 31136, shall contain the following warning:

WARNING

DO NOT OVERHEAT SEALING COMPOUND. STRICTLY OBSERVE RECOMMENDATIONS OF THE SEALING COMPOUND MANUFACTURER RELATIVE TO MELTING AND POURING TEMPERATURES.

THOROUGHLY CLEAN MELTING AND POURING EQUIPMENT WHEN CHANGING FROM SS-S-1401 (RUBBER-ASPHALT) COMPOUNDS TO SS-S-1614 (RUBBER-COAL-TAR) COMPOUNDS, OR VICE VERSA, SINCE THE TWO MATERIALS ARE NOT COMPATIBLE.

USE SOLVENTS, SUCH AS KEROSENE, TO CLEAN KETTLE AND TOOLS AFTER USING SS-S-1401 COMPOUNDS. USE SOLVENTS SUCH AS XYLOL OR TRICHLORETHYLENE AFTER USING SS-S-1614 COMPOUNDS.

3.26 Repair parts and maintenance tools. When specified (see 6.2), repair parts and maintenance tools as are specified shall be furnished.

3.27 Radio interference suppression. When specified (see 6.2), radio interference suppression shall be in accordance with SAE J551c.

3.28 Cleaning, treatment, and painting. Unless otherwise specified (see 6.2), surfaces normally painted in good commercial practice shall be cleaned, treated, and painted as specified herein. The color of the finish coat shall be as specified (see 6.2). Surfaces to be painted shall be cleaned and dried to insure that they are free from contaminants such as oil, grease, welding slag and spatter, loose mill scale, water, dirt, corrosion product, or any other contaminating substances. As soon as practicable after cleaning, and



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before any corrosion product or other contamination can result, the surfaces shall be prepared or treated to insure the adhesion of the coating system. The painting shall consist of at least one coat of primer and one finish coat. The primer shall be applied to a clean, dry surface as soon as practicable after cleaning and treating. Painting shall be with manufacturer's current materials according to manufacturer's current processes and the total dry film thickness shall be not less than 1.5 mils over the entire surface. The paint shall be free from runs, sags, orange peel, or other defects.

3.28.1 Alternate treatment and painting. When specified (see 6.2), the portions of the kettle normally painted shall be cleaned, treated, and painted in accordance with MIL-T-704, type A. The color of the finish coat shall be as specified (see 6.2).

3.29 Workmanship.

3.28.1 Steel fabrication. The steel used in fabrication shall be free from kinks, sharp bends, and other conditions which would be deleterious to the finished product. Manufacturing processes shall not reduce the strength of the steel to a value less than intended by the design. Manufacturing processes shall be done neatly and accurately. All bends shall be made by controlled means to insure uniformity of size and shape.

3.29.2 Bolted connections. Bolt holes shall be accurately punched or drilled and shall have the burrs removed. Washers or lockwashers shall be provided in accordance with good commercial practice, and all bolts, nuts, and screws shall be tight.

3.29.3 Riveted connections. Rivet holes shall be accurately punched or drilled and shall have the burrs removed. Rivets shall be driven with pressure tools and shall completely fill the holes. Rivet heads, when not countersunk or flattened, shall be of approved shape and of uniform size for the same diameter of rivet. Rivet heads shall be full, neatly made, concentric with the rivet holes, and in full contact with the surface of the member.

3.29.4 Welding. Welding procedures shall be in accordance with a nationally recognized welding code. The surface of parts to be welded shall be free from rust, scale, paint, grease, or other foreign matter. Welds shall be of sufficient size and shape to develop the full strength of the parts connected by the welds. Welds shall transmit stress without permanent deformation or failure when the parts connected by the weld are subjected to proof and service loadings.

3.29.5 Castings. All castings shall be sound and free from patching, misplaced coring, warping, or any other defect which reduces the castings ability to perform its intended function.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or purchase order, the contractor may use his own or



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any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.1.1 Components and material inspection. The supplier is responsible for assuring that components and material used are manufactured, tested, and examined in accordance with the referenced specifications and standards.

4.2 Classification of inspection. Inspections are classified as follows:

a. First article inspection (see 4.3).
b. Quality conformance inspection (see 4.4).
c. Packaging inspection (see 4.6).

4.3 First article inspection. First article inspection shall be performed on one kettle when a first article sample is required (see 3.2). This inspection shall include the examination of 4.5.1 and the tests of 4.5.2 through 4.5.2.7. The first article may be a standard production item from the supplier's current inventory provided the kettle meets the requirements of the specification and is representative of the design construction, and manufacturing technique applicable to the remaining kettles to be furnished under the contract. Presence of one or more defects or failure of any test shall be cause for rejection.

4.4 Quality conformance inspection.

4.4.1 Examination. Each production kettle shall be examined as specified in 4.5.1. Presence of one or more defects shall be cause for rejection.

4.4.2 Test. Each production kettle shall be tested as specified in 4.5.3. Failure of this test shall be cause for rejection.

4.5 Inspection procedure.

4.5.1 Examination. The kettle shall be examined for the following defects:

| | |
|---|---|
| 101 | Material not as specified. |
| 102 | Oil compartment not as specified. |
| 103 | Melting compartment not as specified. |
| 104 | Trailer not as specified. |
| 105 | Lunette and safety chains not as specified. |
| 106 | Support jacks not as specified. |
| 107 | Lubrication fittings not accessible or not as specified. |
| 108 | Cleaning, treatment, painting, or stenciling not as specified. |
| 109 | Technical publications not as specified. |
| 110 | Identification and other marking or operational and safety instructions missing, illegible, or not as specified. |
| 111 | Repair parts and maintenance tools not as specified. |
| 112 | Engine and engine components not as specified. |
| 113 | Burner(s) not as specified. |
| 114 | Pumping system not as specified. |



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115 Discharge hose, sealer bar, or sealer tips not as specified.
116 Agitator not as specified.
117 Battery and battery box not as specified.
118 Controls not accessible or not identified.
119 Evidence of leaks at gaskets, seals, or joints (including leaks between the heat-transfer oil compartment and the melting compartment).
120 Workmanship not as specified.

4.5.2 Tests.

4.5.2.1 Road. Fill the kettle to capacity with heat-transfer oil and fuel (two cylinders filled with LPG gas for type I, III, and IV) and tow the kettle for 4 hours on paved roads at an average rate of 25 mph. At the end of this period, examine the kettle. Evidence of leaks, misalinement of parts or components, weld failure, or damage to any part or component, of the kettle shall constitute failure of this test.

4.5.2.2 Initial melt. Charge the melting compartment (see 3.7) with joint-sealing compound conforming to SS-S-1401 or SS-S-1614. Ignite the burners and operate for 3 hours or until the sealing compound reaches a temperature of 415°F. A sealing compound temperature of less than 415°F after 3 hours of operation shall constitute failure of this test.

4.5.2.3 Melting. After completion of the test specified in 4.5.2.2, operate the kettle for 2 hours under simulated operating conditions. Inability of the kettle to maintain the sealing compound temperature to the temperature recommended by the manufacturer and at the rate specified in 3.7 or inability of the pumping system (except type IV) to perform as specified in 3.7 shall constitute failure of this test.

4.5.2.4 Temperature control. Throughout the tests specified in 4.5.2.2 and 4.5.2.3 record the temperature of the heat-transfer oil at 15 minute intervals. Inability of the kettle to maintain the heat-transfer oil at the preselected temperature with ±10°F for the type I, III, and IV or within ±25°F for the type II shall constitute failure of this test.

4.5.2.5 Lifting attachment. When equipped with lifting attachments, suspend the empty kettle by the lifting attachments using multiple slings. Determine the load imposed on each attachment with the kettle suspended. Subject each attachment to a load 2-1/2 times greater than the load imposed on the attachment when the kettle was suspended. Nonconformance to 3.20 or weld failure or permanent deformation to any attachment or structural member shall be cause for rejection.

4.5.2.6 Tiedown attachment. When equipped with tiedown attachments, subject each tiedown attachment to the proportionate share of the forward, rearward, upward, and sideward loads specified in 3.21. Load measurements shall be made with a dynamometer or other accurate measuring device. Nonconformance to 3.21 or weld failure or permanent deformation to any attachment or structural member shall be cause for rejection.



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4.5.2.7 Radio interference suppression. To determine conformance to 3.27, electromagnetic radiation shall be measured in accordance with SAE J551. When suppressed to conform to 3.27, the manufacturer may furnish a certification in lieu of the test that the kettles meet the requirements, together with a list of the suppression devices installed. The list shall be sufficiently detailed to allow visual determination that the devices are installed. Failure to comply shall be cause for rejection.

4.5.3 Functional test. Each kettle shall be serviced in accordance with the manufacturer's recommendation prior to testing. Fill each kettle to the operating level with heat-transfer oil. Ignite the burner and operate for a period of time necessary to determine that the burner and all burner controls function properly. On types I, II, and III kettles, determine that the engine and all its controls operate properly. Oil leaks, malfunction of burner and burner controls, or malfunction of the engine and engine controls which cannot be corrected by minor adjustment shall constitute failure of this test.

4.6 Packaging inspection. The inspection of the preservation, packing, and marking shall be in accordance with the requirements of Section 4 of MIL-D-771. The inspection shall consist of the quality conformance inspection; and, when specified (see 6.2), a preproduction pack shall be furnished for examination and test within the time frame required (see 6.2).

## 5. PACKAGING

5.1 Preservation, packing, and marking. Preservation, packing, and marking shall be in accordance with the requirements of MIL-D-771 with the level of preservation and the level of packing as specified (see 6.2).

## 6. NOTES

6.1 Intended use. The kettles are intended to melt and dispense rubber-asphalt and rubber-coal-tar compounds for sealing expansion joints in concrete roadways and airfields.

6.2 Ordering data. Acquisition documents should specify the following:

a. Title, number, and date of this specification.
b. Type and size of kettle to be furnished (see 1.2).
c. When a first article is required for inspection and approval (see 3.2).
d. When a cover is required (see 3.13).
e. When electrolyte shall be furnished (see 3.15.10.1).
f. When the battery shall conform to MIL-B-23272 (see 3.15.10.1).
g. When lifting attachments are required (see 3.20).
h. When tiedown attachments are required (see 3.21).
i. When material for identification and transportation-data plates shall be other than as specified (see 3.22).
j. When stenciling is required (see 3.23).
k. When technical publications required (see 3.24).
l. When repair parts and maintenance tools shall be furnished (see 3.26).
m. When radio interference suppression is required (see 3.27).



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n. When cleaning, treatment, and painting shall be commercial standard (see 3.28).
o. Finish color coat required (see 3.28).
p. When alternate treatment and painting (MIL-T-704) is required (see 3.28.1).
q. Alternate treatment and painting finish color coat required (see 3.28.1).
r. When a preproduction pack is required and time frame required for submission (see 4.6).
s. Level of preservation and level of packing required (see 5.1).

6.3 Data requirements. When this specification is used in an acquisition which incorporates a DD Form 1423 and invokes the provisions of paragraph 7-104.9(n) of the Defense Acquisition Regulations (DAR), the data requirements will be developed as specified by an approved Data Item Description (DD Form 1664) and delivered in accordance with the Contract Data Requirements List (DD Form 1423) incorporated into the contract. When the provisions of DAR 7-104.9(n) are not invoked, the data shall be delivered in accordance with the contract requirements.

6.4 First article. When a first article inspection is required, the item will be tested and should be a first article sample or it may be a standard production item from the contractor's current inventory as specified in 4.3. The first article should consist of one kettle. The contracting officer should include specific instructions in acquisition documents regarding arrangements for examination, test, and approval of the first article.

6.5 Changes from previous issue. Asterisks are not used in this revision to identify changes with respect to the previous issue, due to the extensiveness of the changes.

Custodian:
Navy - YD
Air Force - 99

User activity:
Army - CE

Preparing activity:
Air Force - 99

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DOCUMENT IDENTIFIER *(Number)* AND TITLE
MIL-K-4877C, Kettles, Heating, Bitumen, Joint-Sealer Trailer Mounted

NAME OF ORGANIZATION AND ADDRESS OF SUBMITTER

☐ VENDOR ☐ USER ☐ MANUFACTURER

1. ☐ HAS ANY PART OF THE DOCUMENT CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE? ☐ IS ANY PART OF IT TOO RIGID, RESTRICTIVE, LOOSE OR AMBIGUOUS? PLEASE EXPLAIN BELOW.

A. GIVE PARAGRAPH NUMBER AND WORDING

B. RECOMMENDED WORDING CHANGE

C. REASON FOR RECOMMENDED CHANGE(S)

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